

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

Practical No. 2 (CSV)

Aim: Working with Data Sources and Data Preparation in Tableau

- A. Connect to different data sources (Excel, CSV, Database)
- B. Create and use data extracts vs live connections
- C. Manage metadata (rename, hide, change data types)
- D. Perform joins (inner, left, right) on multiple tables
- E. Apply data blending between two sources
- F. Implement different types of filters (dimension, measure, date)
- G. Optimize performance using extracts and filtering

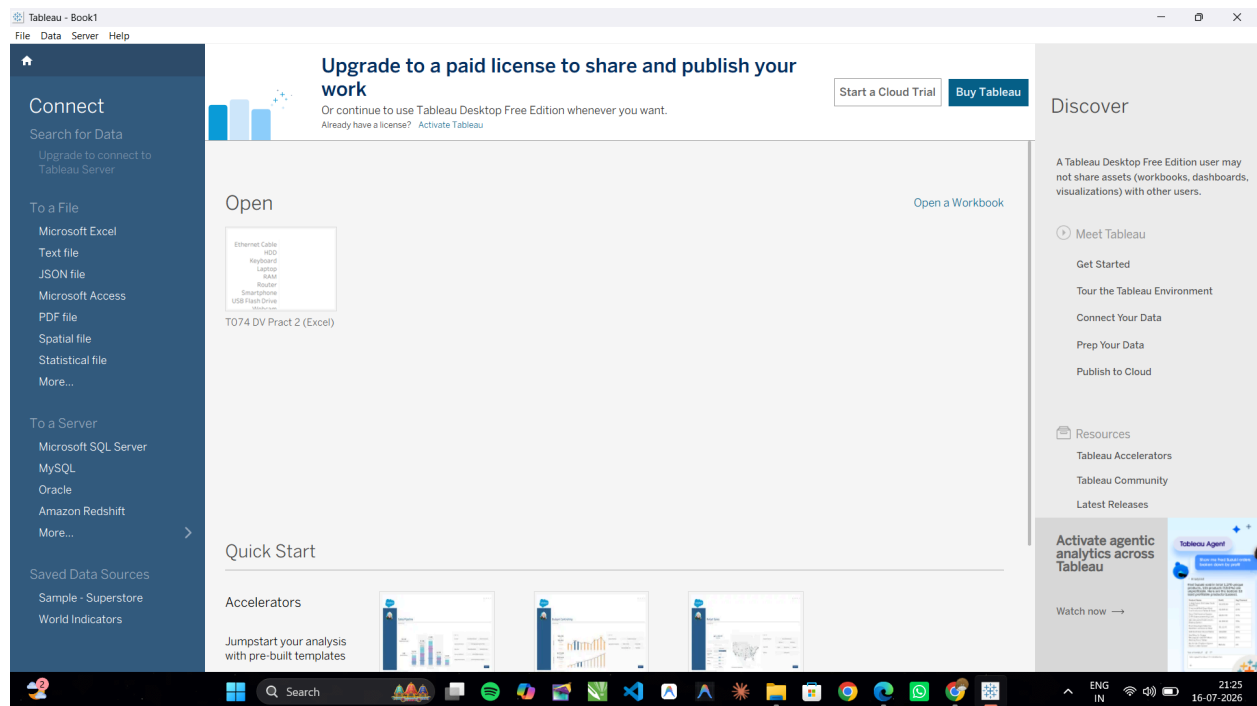
Dataset: Inventory-Records-Sample-Data.xlsx

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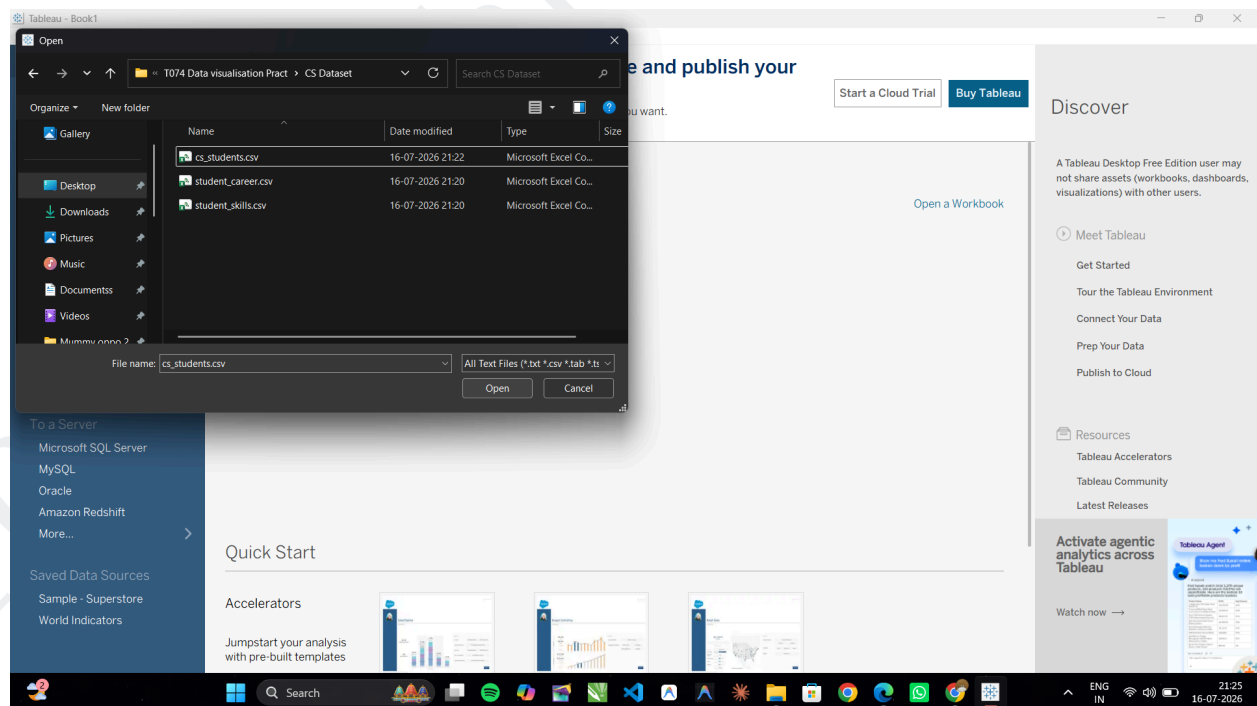
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Step A: Connect to different data sources (Excel, CSV, Database)

1. Launch **Tableau** Desktop on your machine



2. In the **Connect** pane, click **Text File** Browse and select **cs_students.csv** Click **Open** to load the dataset.



Krishna Nandkumar Chaudhary

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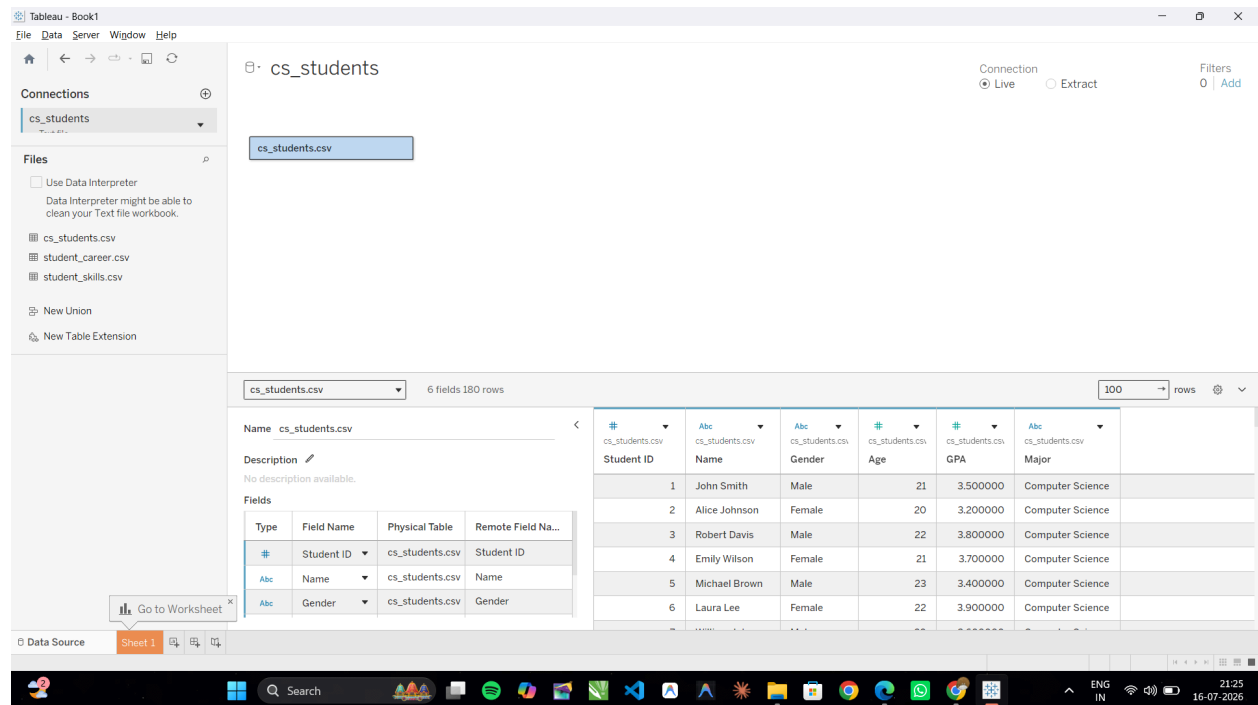


Tableau - Book1

File Data Server Window Help

Connections

- cs_students

Files

- Use Data Interpreter
- cs_students.csv
- student_career.csv
- student_skills.csv
- New Union
- New Table Extension

cs_students

cs_students.csv

Connection: Live Extract Filters: 0 Add

cs_students.csv 6 fields 180 rows 100 rows

Name cs_students.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field Na...
#	Student ID	cs_students.csv	Student ID
Abc	Name	cs_students.csv	Name
Abc	Gender	cs_students.csv	Gender

#	Abc	Abc	#	#	Abc
cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv
Student ID	Name	Gender	Age	GPA	Major
1	John Smith	Male	21	3.500000	Computer Science
2	Alice Johnson	Female	20	3.200000	Computer Science
3	Robert Davis	Male	22	3.800000	Computer Science
4	Emily Wilson	Female	21	3.700000	Computer Science
5	Michael Brown	Male	23	3.400000	Computer Science
6	Laura Lee	Female	22	3.900000	Computer Science

Go to Worksheet

Data Source Sheet 1

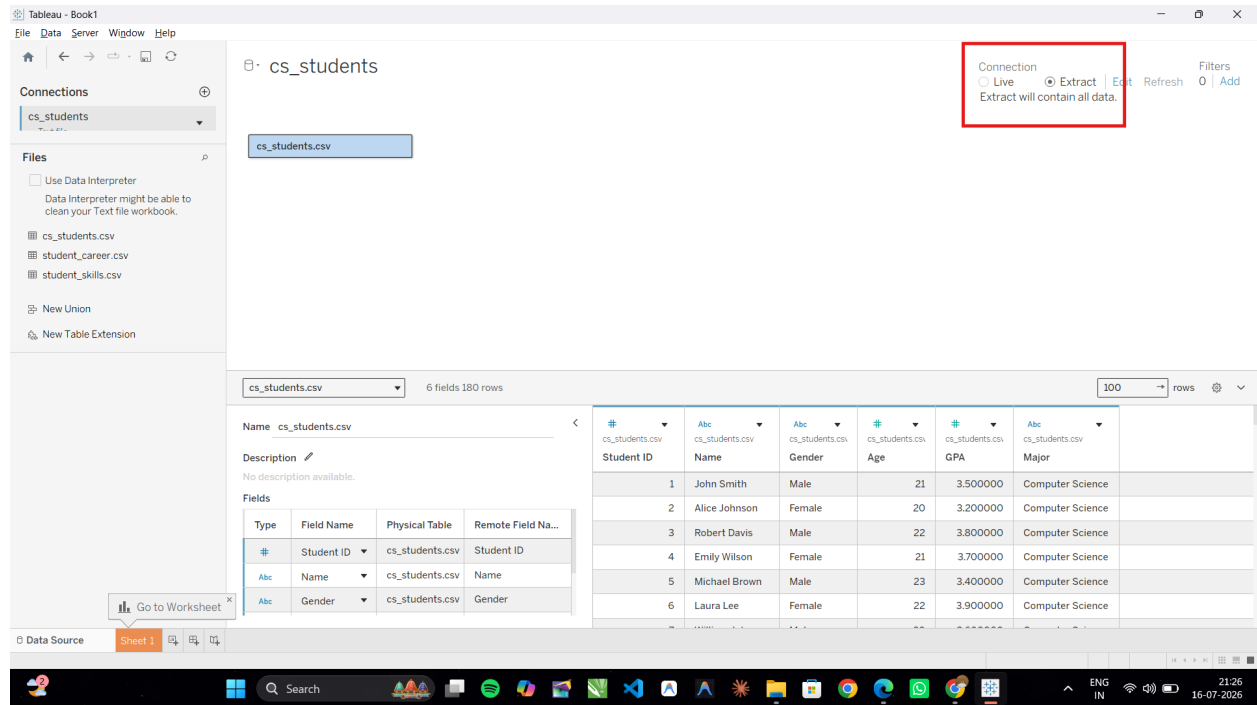
21:25 16-07-2026

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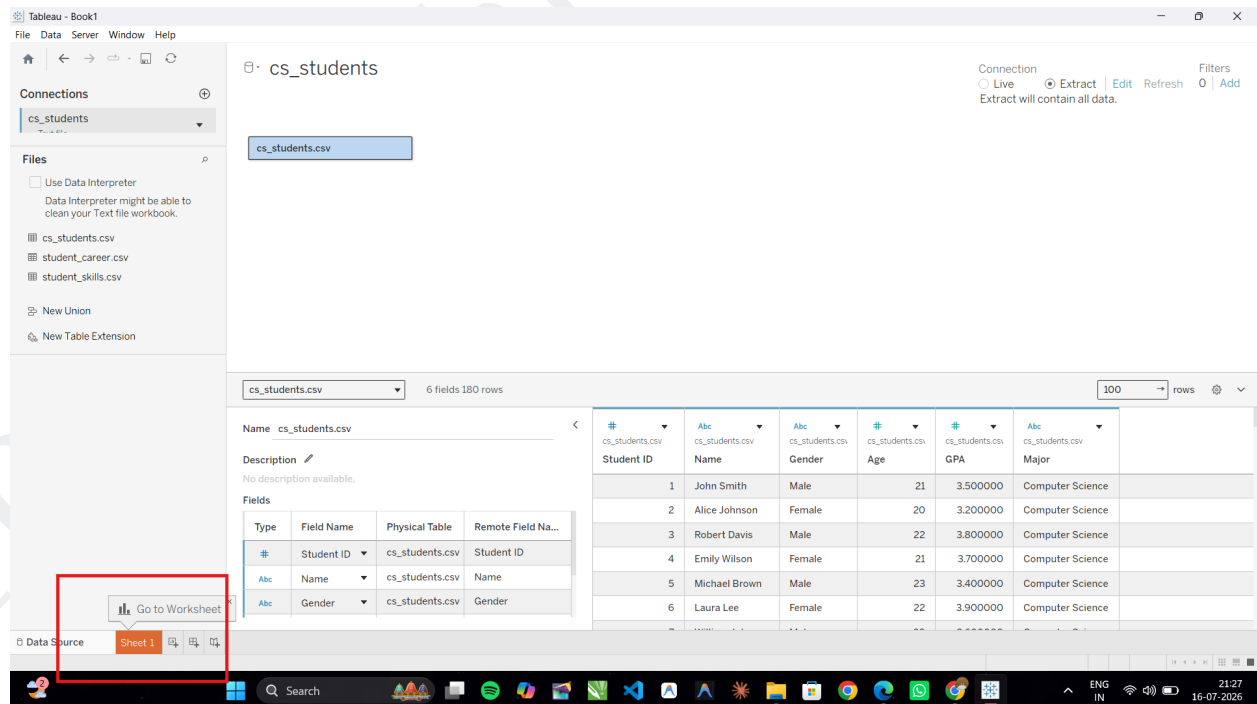
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Step B: Creating Data Extracts vs. Live Connections

1. In the **Data Source** page, select **Extract** instead of **Live**.



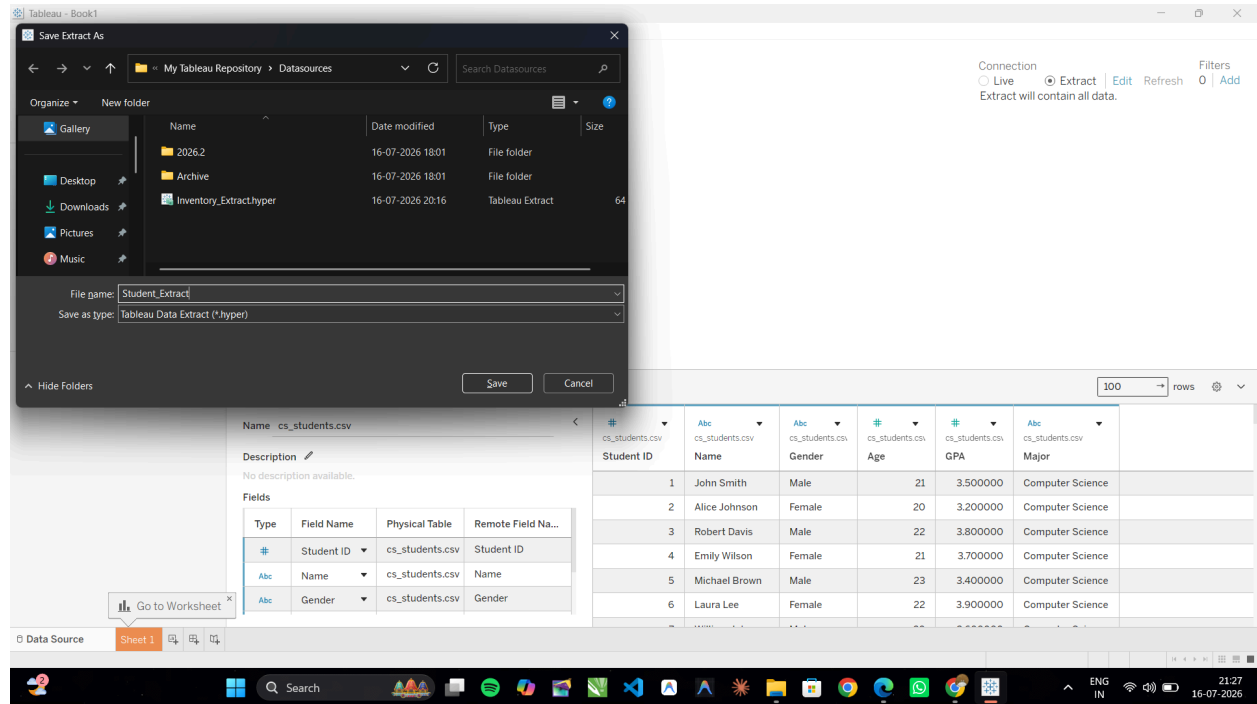
2. Click **Sheet 1**.



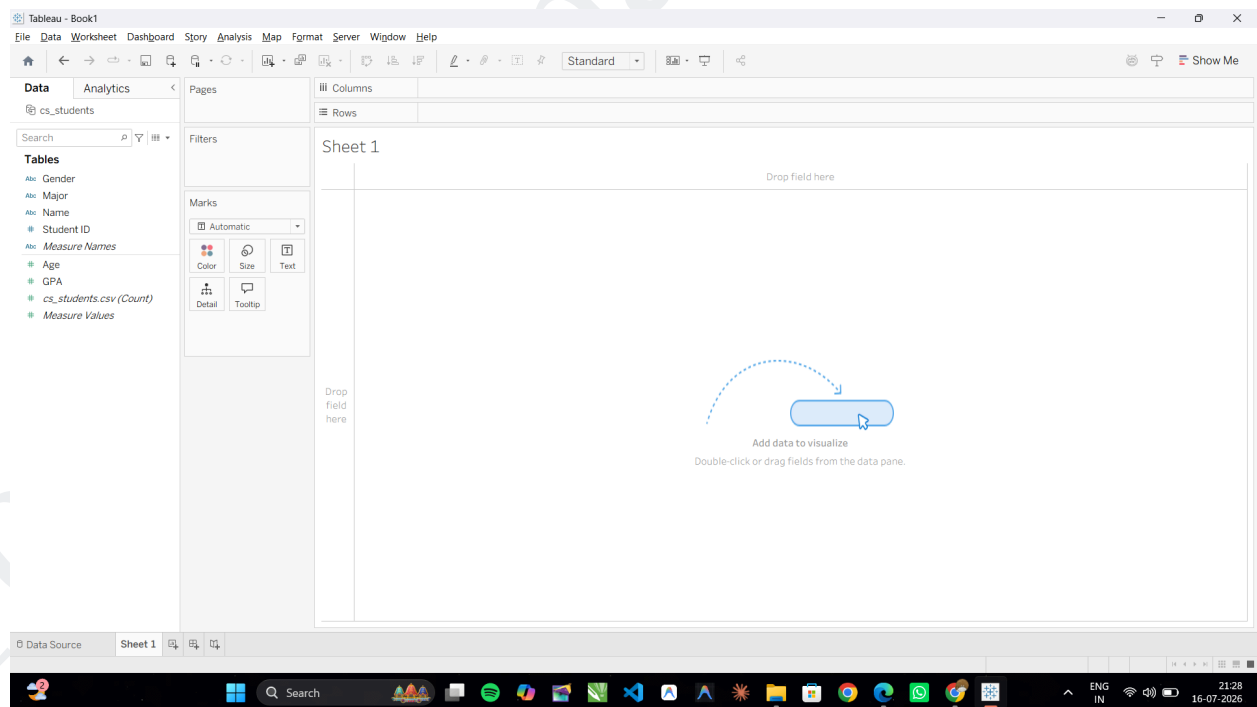
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3. Save the extract as **Student_Extract.hyper**.



4. The extract is created and the worksheet opens.

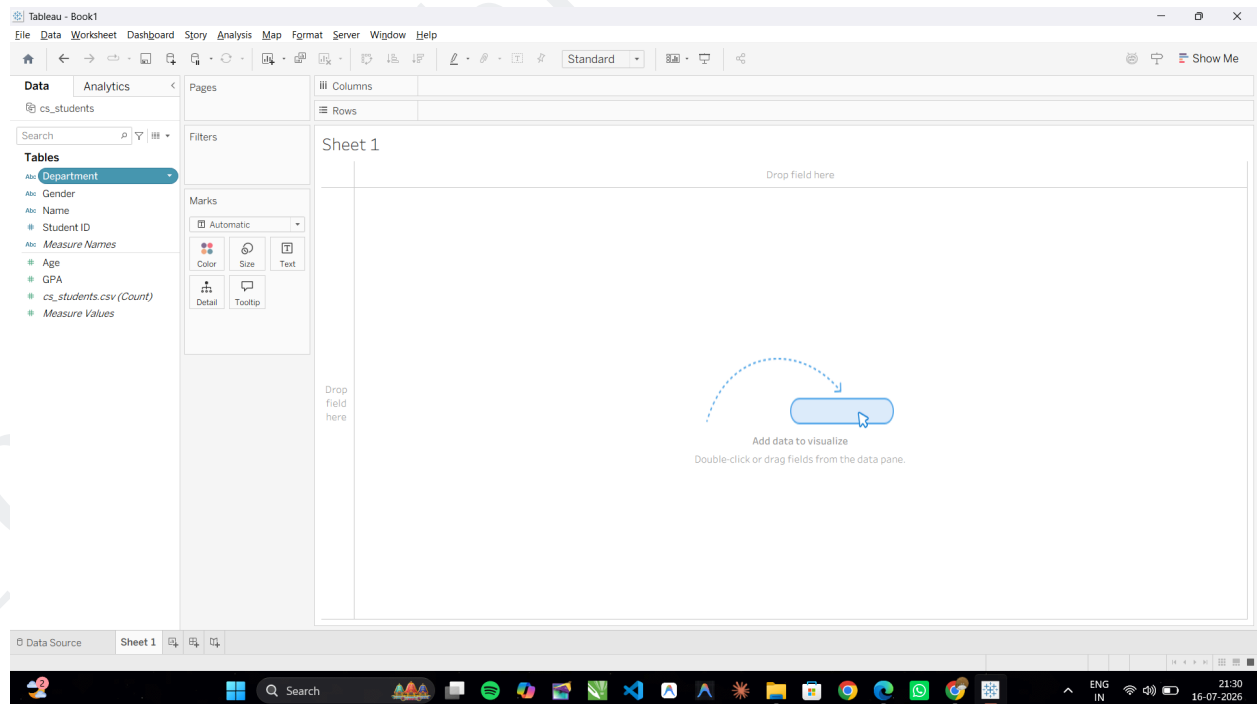
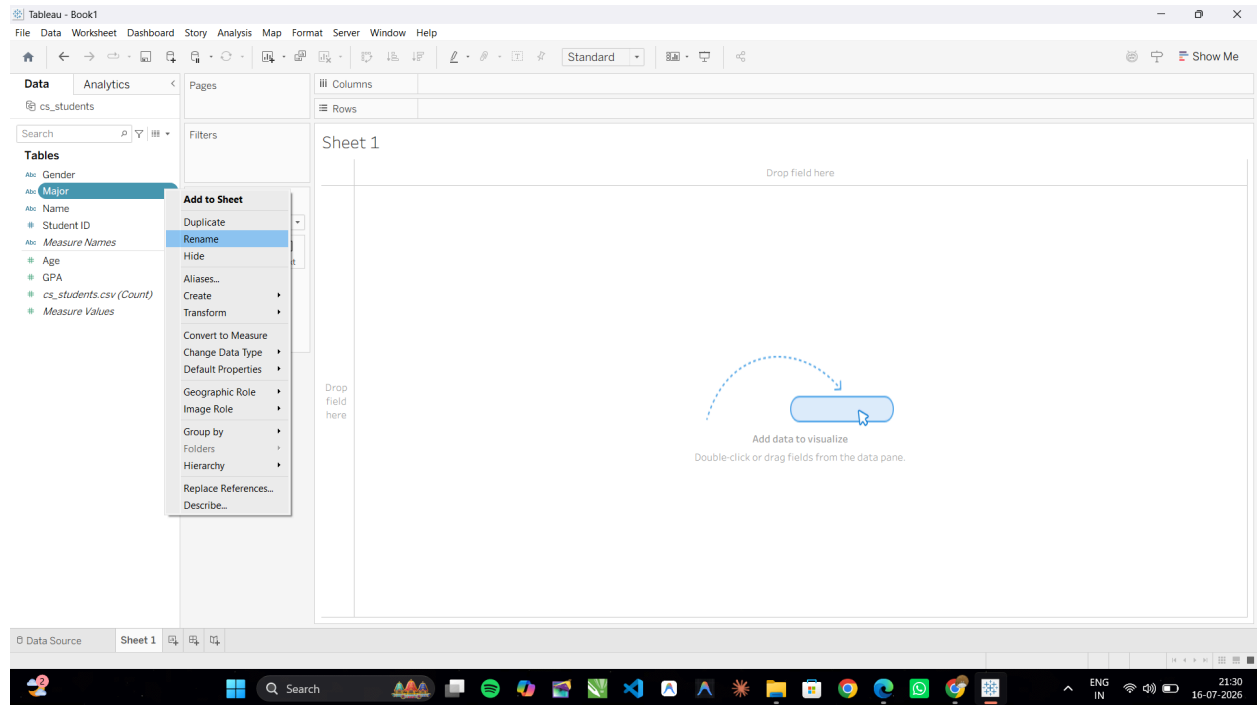


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Step C: Managing Operational Metadata

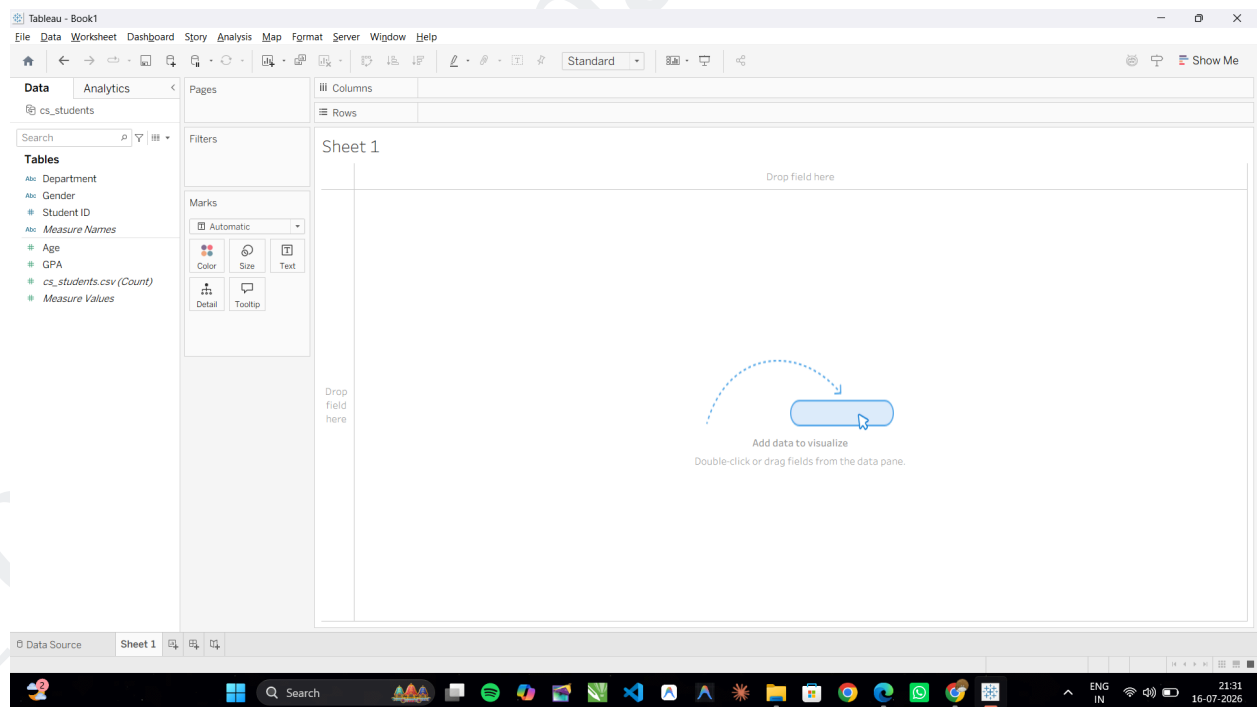
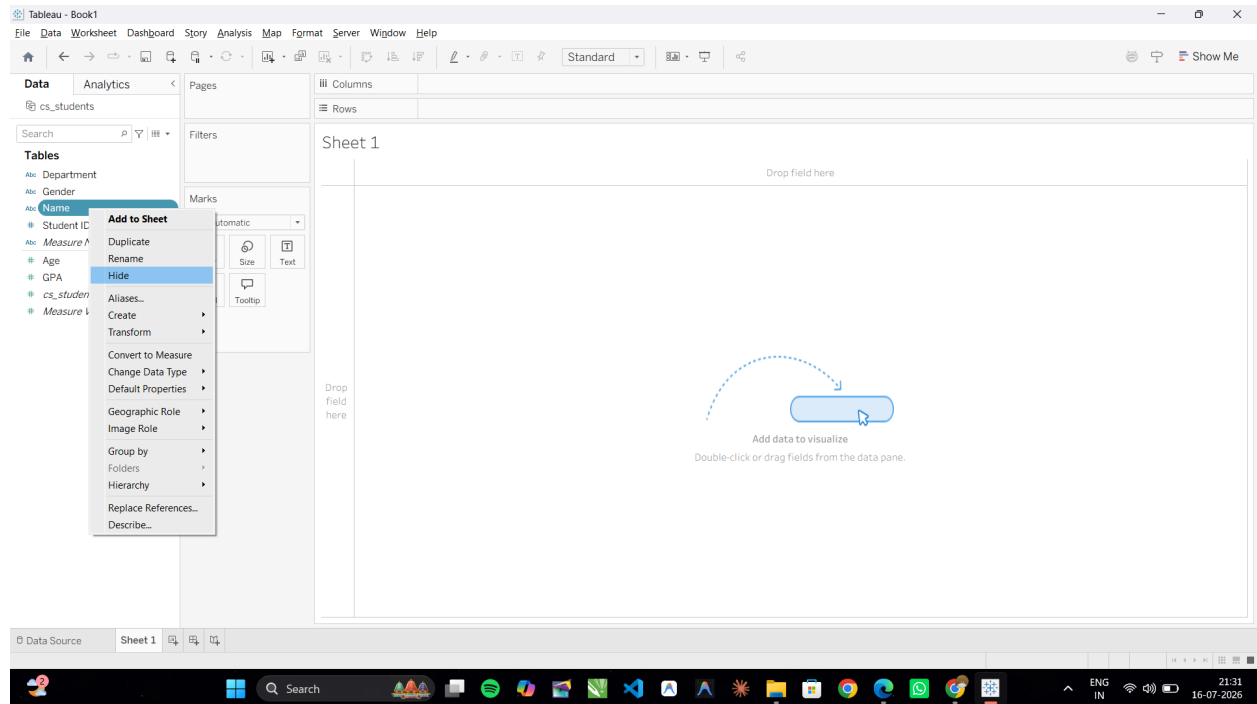
1. Rename a Field: Right-click the **Major** field Select **Rename**, Rename it to **Department**.



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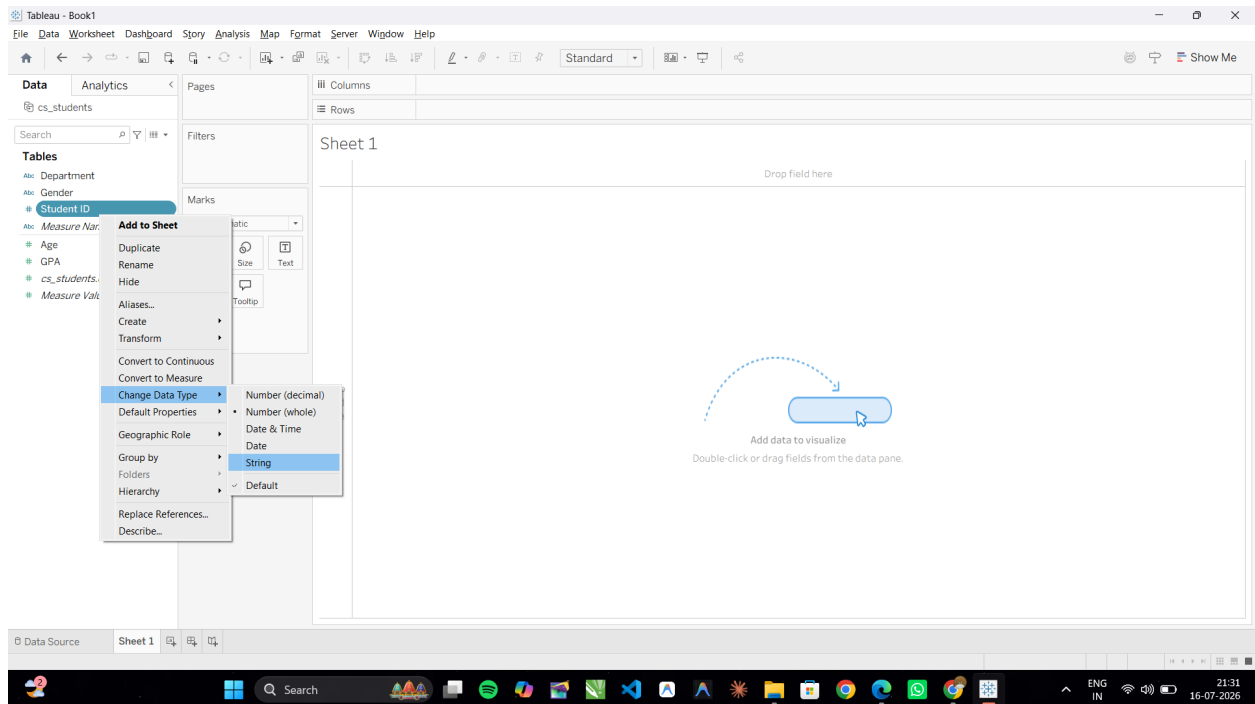
2. Hide a Field: Right-click the **Name** field Select **Hide**.



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3. Change a Data Type: Click the data type icon next to **Student ID** and Change the data type from **Number** to **String**.



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Step D: Performing Relational Joins on Multiple Tables

1. Open the **Data Source** page and Double-click **cs_students.csv** to enter the **Physical Layer**.

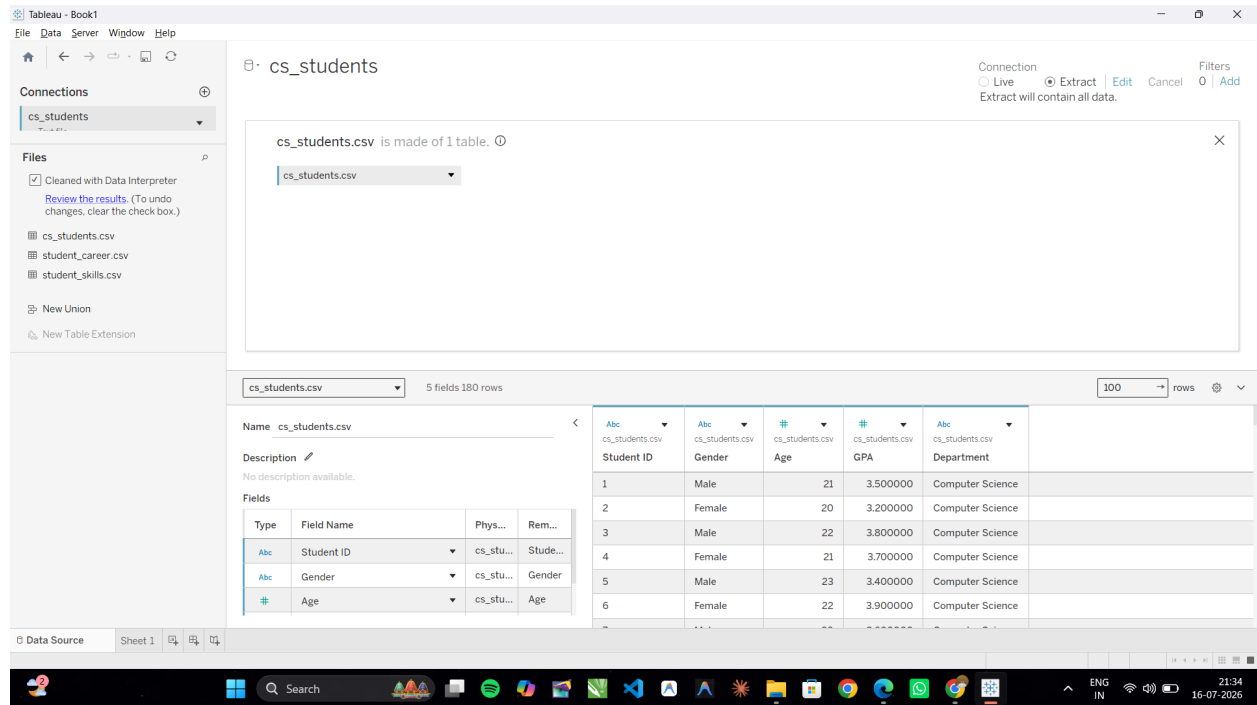


Tableau - Book1

File Data Server Window Help

Connections

cs_students

Files

☒ Cleaned with Data Interpreter
[Review the results.](#) (To undo changes, clear the check box.)

cs_students.csv
student_career.csv
student_skills.csv

New Union
New Table Extension

cs_students

Connection: ☐ Live ☒ Extract | [Edit](#) | Cancel | Filters: 0 | [Add](#)
Extract will contain all data.

cs_students.csv is made of 1 table.

cs_students.csv

cs_students.csv 5 fields 180 rows 100 rows

Name cs_students.csv

Description No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Student ID	cs_stu...	Stude...
Abc	Gender	cs_stu...	Gender
#	Age	cs_stu...	Age

Abc	Abc	#	#	Abc
cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv
Student ID	Gender	Age	GPA	Department
1	Male	21	3.500000	Computer Science
2	Female	20	3.200000	Computer Science
3	Male	22	3.800000	Computer Science
4	Female	21	3.700000	Computer Science
5	Male	23	3.400000	Computer Science
6	Female	22	3.900000	Computer Science

2. Add **student_skills.csv**.

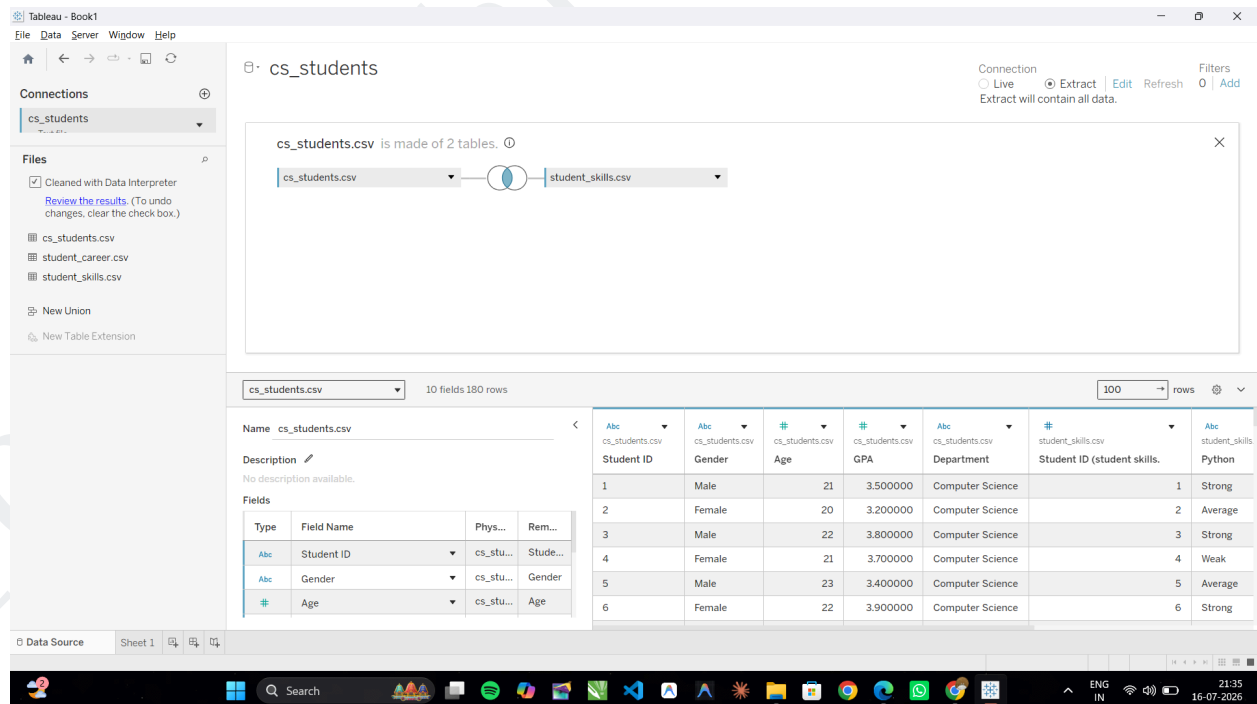


Tableau - Book1

File Data Server Window Help

Connections

cs_students

Files

☒ Cleaned with Data Interpreter
[Review the results.](#) (To undo changes, clear the check box.)

cs_students.csv
student_career.csv
student_skills.csv

New Union
New Table Extension

cs_students

Connection: ☐ Live ☒ Extract | [Edit](#) | Refresh | Filters: 0 | [Add](#)
Extract will contain all data.

cs_students.csv is made of 2 tables.

cs_students.csv student_skills.csv

cs_students.csv 10 fields 180 rows 100 rows

Name cs_students.csv

Description No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Student ID	cs_stu...	Stude...
Abc	Gender	cs_stu...	Gender
#	Age	cs_stu...	Age

Abc	Abc	#	#	Abc	#	Abc
cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv	student_skills.csv	student_skills
Student ID	Gender	Age	GPA	Department	Student ID (student skills)	Python
1	Male	21	3.500000	Computer Science	1	Strong
2	Female	20	3.200000	Computer Science	2	Average
3	Male	22	3.800000	Computer Science	3	Strong
4	Female	21	3.700000	Computer Science	4	Weak
5	Male	23	3.400000	Computer Science	5	Average
6	Female	22	3.900000	Computer Science	6	Strong

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4. Click the **Join (Venn Diagram)** icon.

Test the following join types:

- Inner Join
- Left Join
- Right Join
- Full Outer Join

Keep the join type as **Left Join**.

The screenshot shows the Tableau Desktop interface. In the Connections pane, 'cs_students' is selected. The main workspace displays a Venn diagram for a join between 'cs_students.csv' and 'student_skills.csv'. The 'Join' dialog box is open, showing 'Left' as the selected join type. The 'Data Source' is 'cs_students.csv' and the 'Table' is 'student_skills.csv'. The 'Student ID' field is selected for the join. Below the dialog, the 'cs_students.csv' data source is shown with 10 fields and 180 rows. The Fields shelf contains 'Student ID', 'Gender', and 'Age'. The data preview table is displayed below the Fields shelf.

Type	Field Name	Phys...	Rem...
Abc	Student ID	cs_stu...	Stude...
Abc	Gender	cs_stu...	Gender
#	Age	cs_stu...	Age

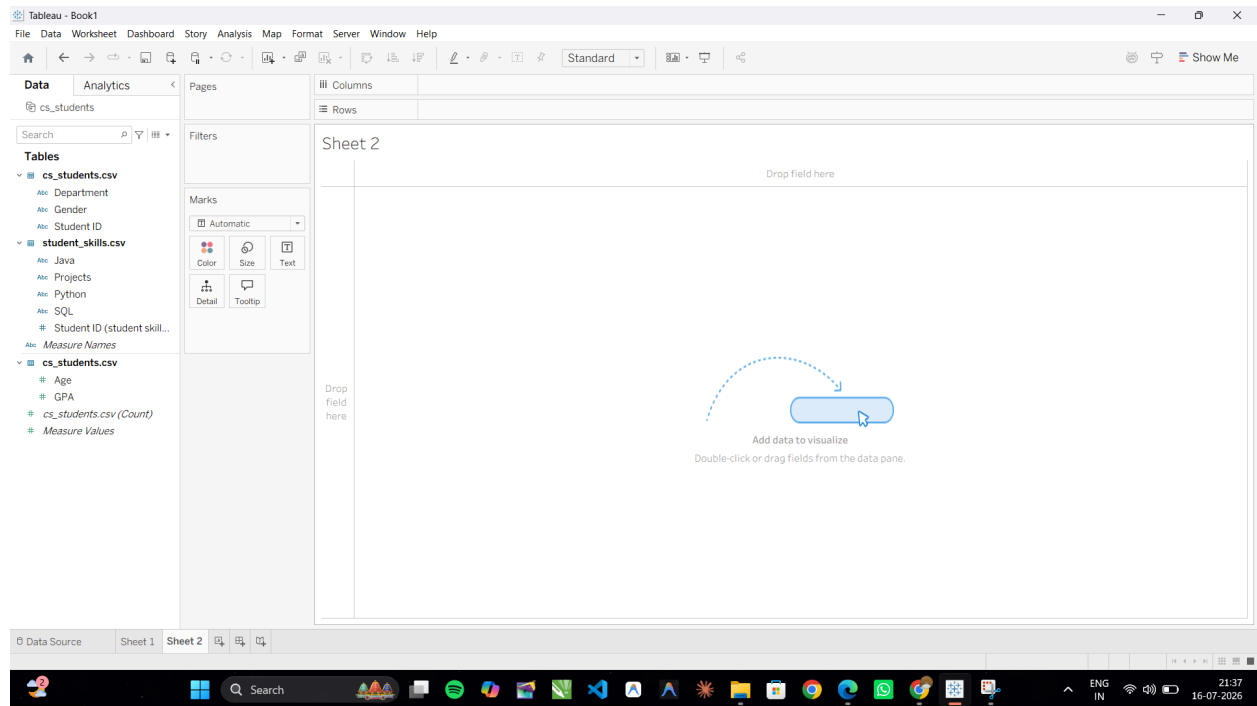
Abc	Abc	#	#	Abc	#	Abc
cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv	cs_students.csv	student_skills.csv	student_skills
Student ID	Gender	Age	GPA	Department	Student ID (student skills)	Python
1	Male	21	3.500000	Computer Science	1	Strong
2	Female	20	3.200000	Computer Science	2	Average
3	Male	22	3.800000	Computer Science	3	Strong
4	Female	21	3.700000	Computer Science	4	Weak
5	Male	23	3.400000	Computer Science	5	Average
6	Female	22	3.900000	Computer Science	6	Strong

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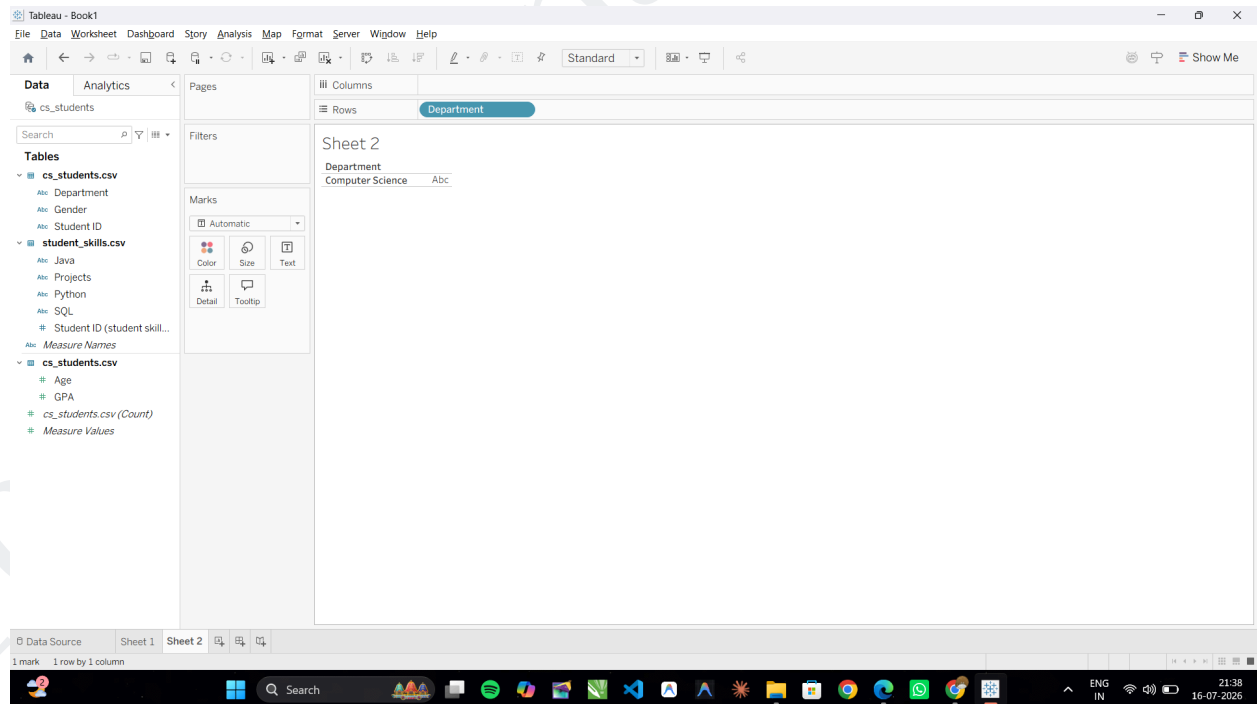
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Step E: Apply Data Blending

1. Create a new worksheet.



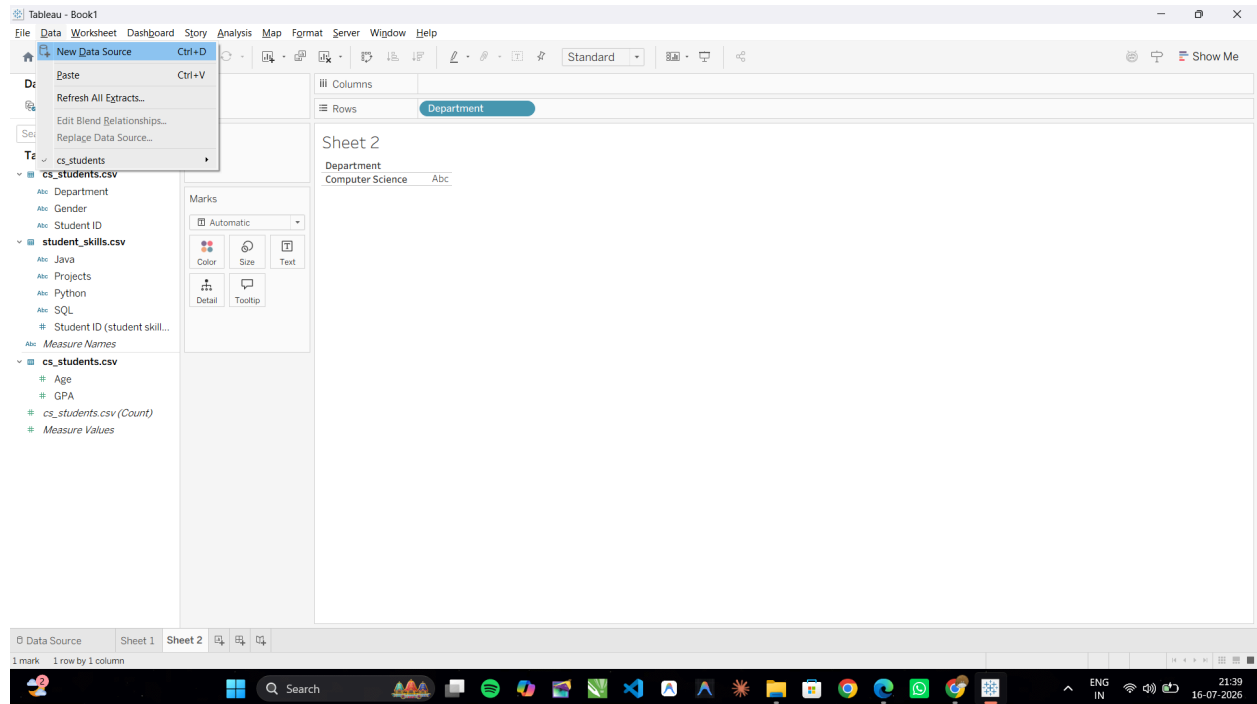
2. Drag Department to the Rows shelf.



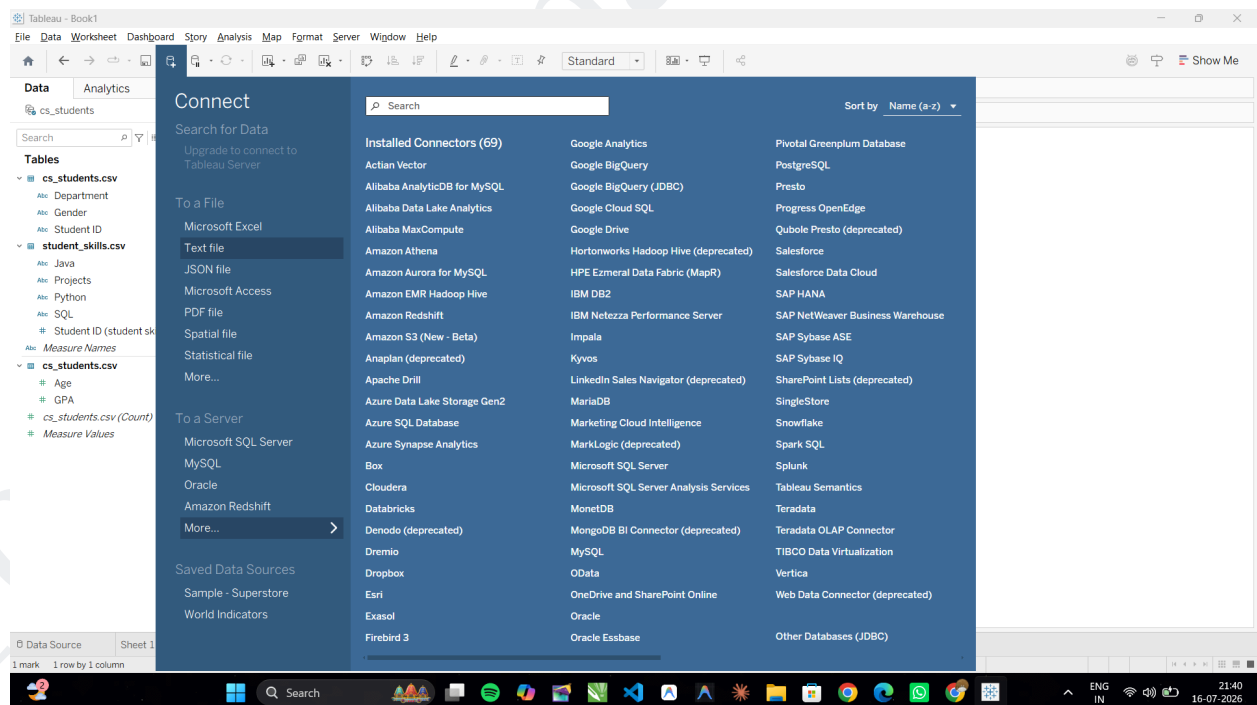
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3. Go to Data → New Data Source.



4. Connect student_career.csv.



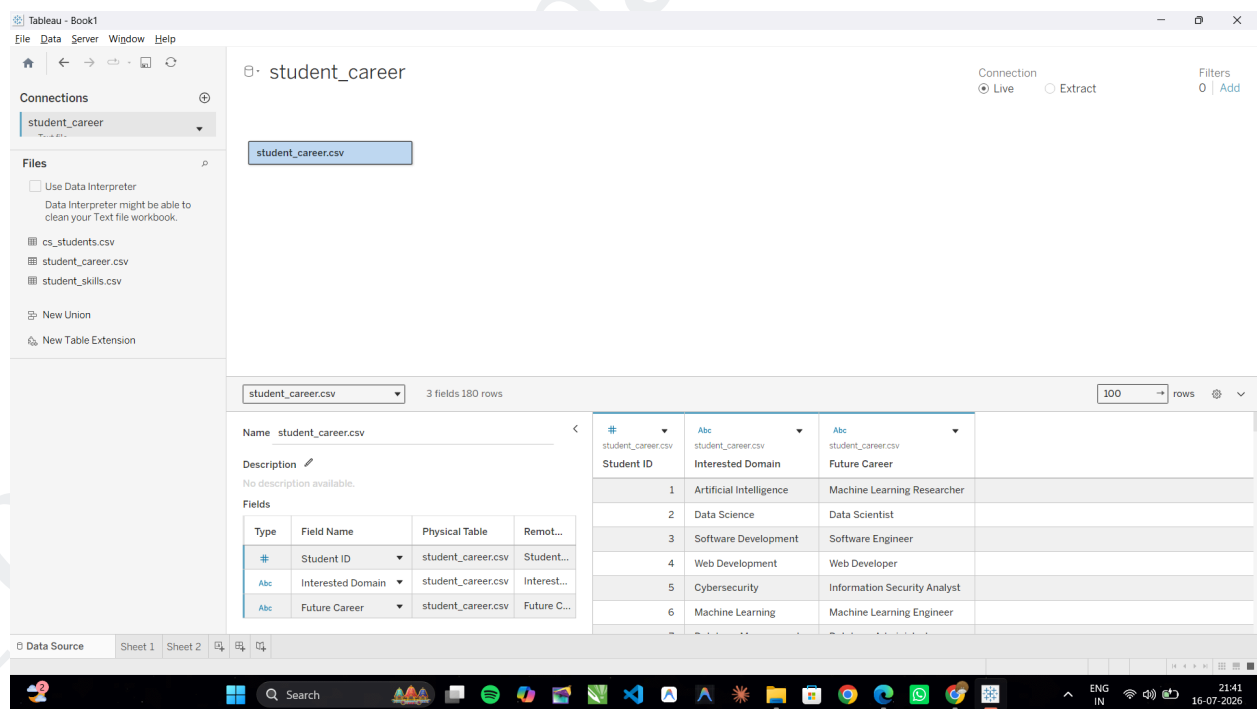
The screenshot shows the Tableau Desktop interface. In the foreground, the 'Open' dialog box is open, displaying a file named 'student_career.csv' selected. The 'Connect to a Server' menu is also open, showing a list of data sources. The background shows a Tableau worksheet with a table of data.

Open Dialog Box:

- File name: student_career.csv
- All Text Files (*.txt; *.tab; *.t)
- Buttons: Open, Cancel

Connect to a Server Menu:

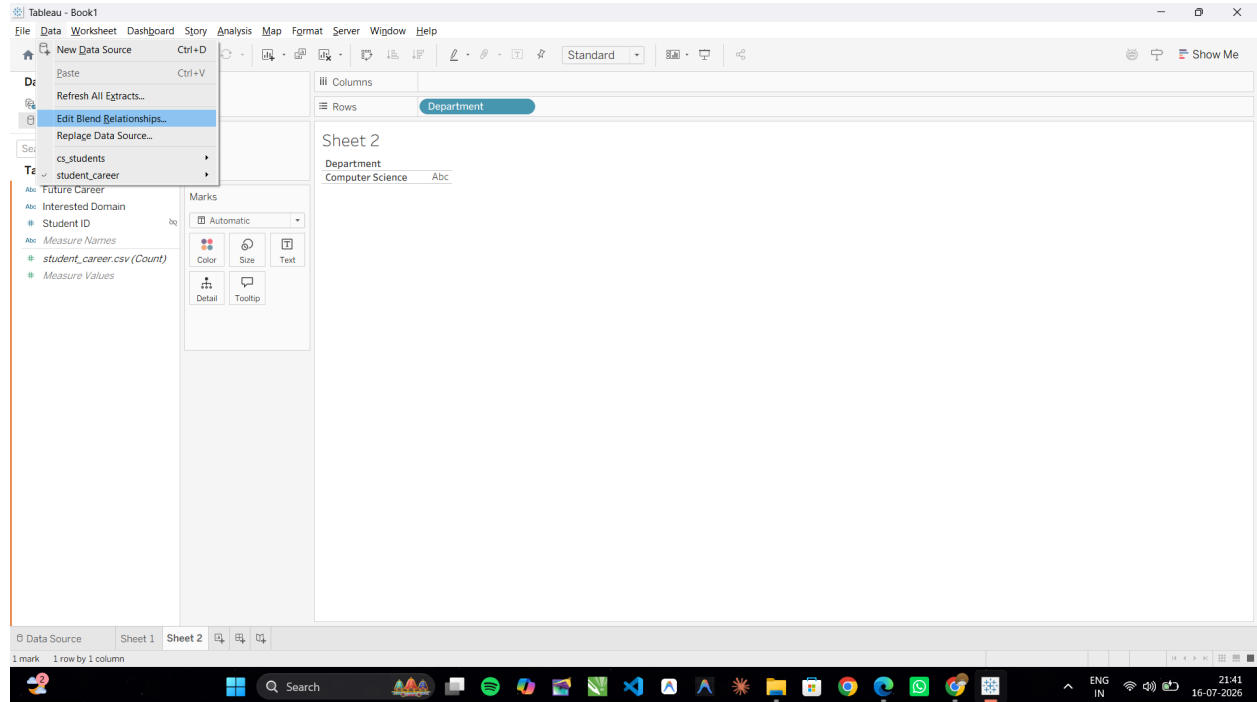
- Microsoft SQL Server
- Oracle
- Amazon Redshift
- More...
- Azure SQL Database
- Azure Synapse Analytics
- Box
- Cloudera
- Databricks
- Denodo (deprecated)
- Dremio
- Dropbox
- Esri
- Exasol
- Firebird 3
- Marketing Cloud Intelligence
- MarkLogic (deprecated)
- Microsoft SQL Server
- Microsoft SQL Server Analysis Services
- MonetDB
- MongoDB BI Connector (deprecated)
- MySQL
- OData
- OneDrive and SharePoint Online
- Oracle
- Oracle Essbase
- Pivotal Greenplum Database
- PostgreSQL
- Presto
- Progress OpenEdge
- Qubole Presto (deprecated)
- Salesforce
- Salesforce Data Cloud
- SAP HANA
- SAP NetWeaver Business Warehouse
- SAP Sybase ASE
- SAP Sybase IQ
- SharePoint Lists (deprecated)
- SingleStore
- Snowflake
- Spark SQL
- Splunk
- Tableau Semantics
- Teradata
- Teradata OLAP Connector
- TIBCO Data Virtualization
- Vertica
- Web Data Connector (deprecated)
- Other Databases (JDBC)



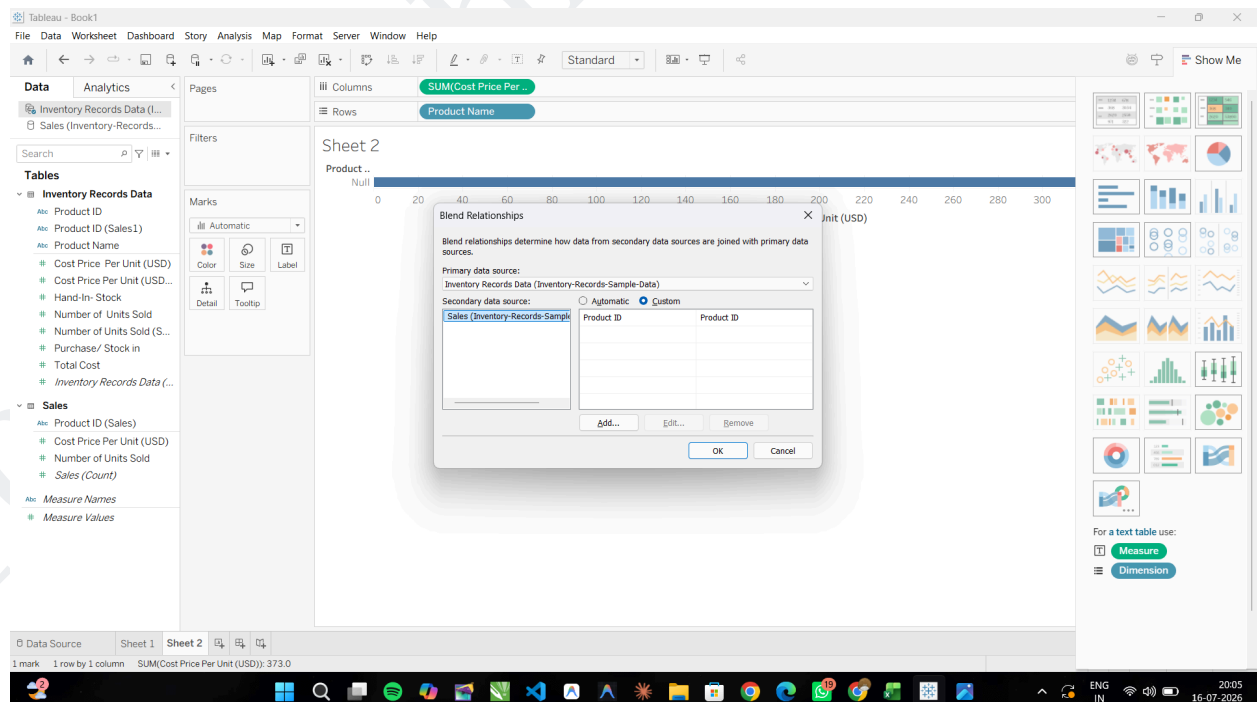
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5. Go to **Data** → **Edit Blend Relationships**.



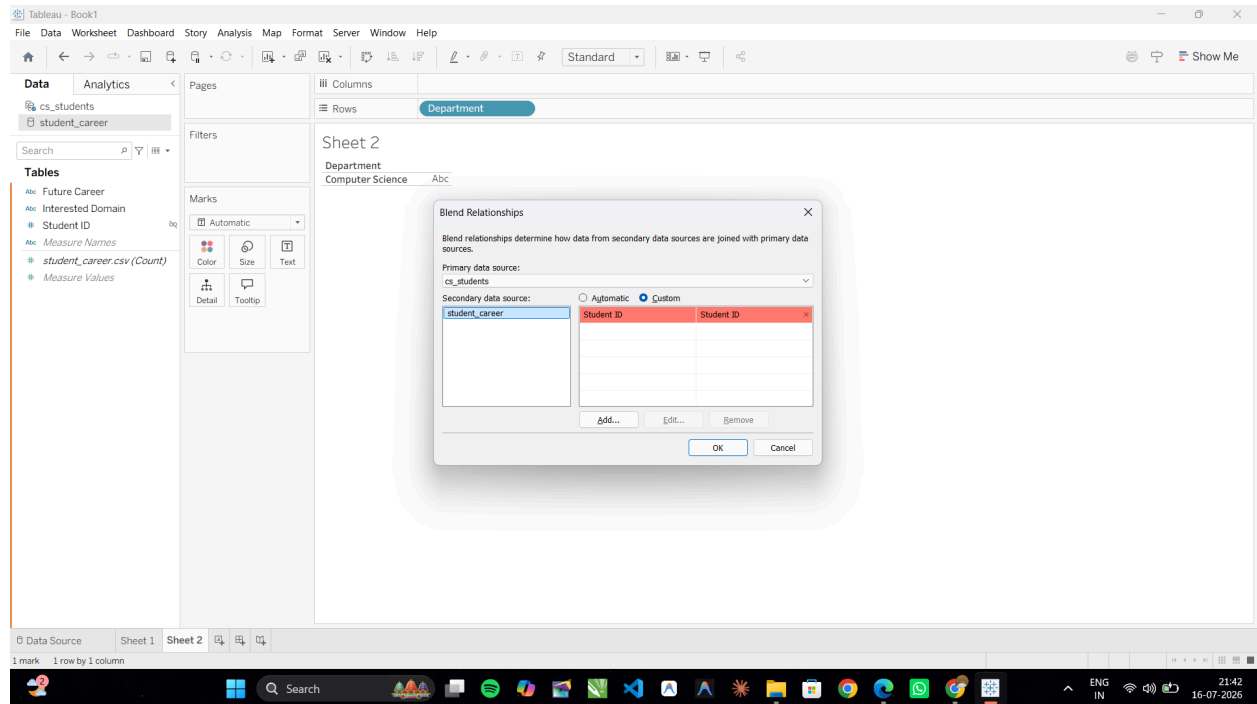
6. Select: Primary Source: **cs_students**, Secondary Source: **student_career**



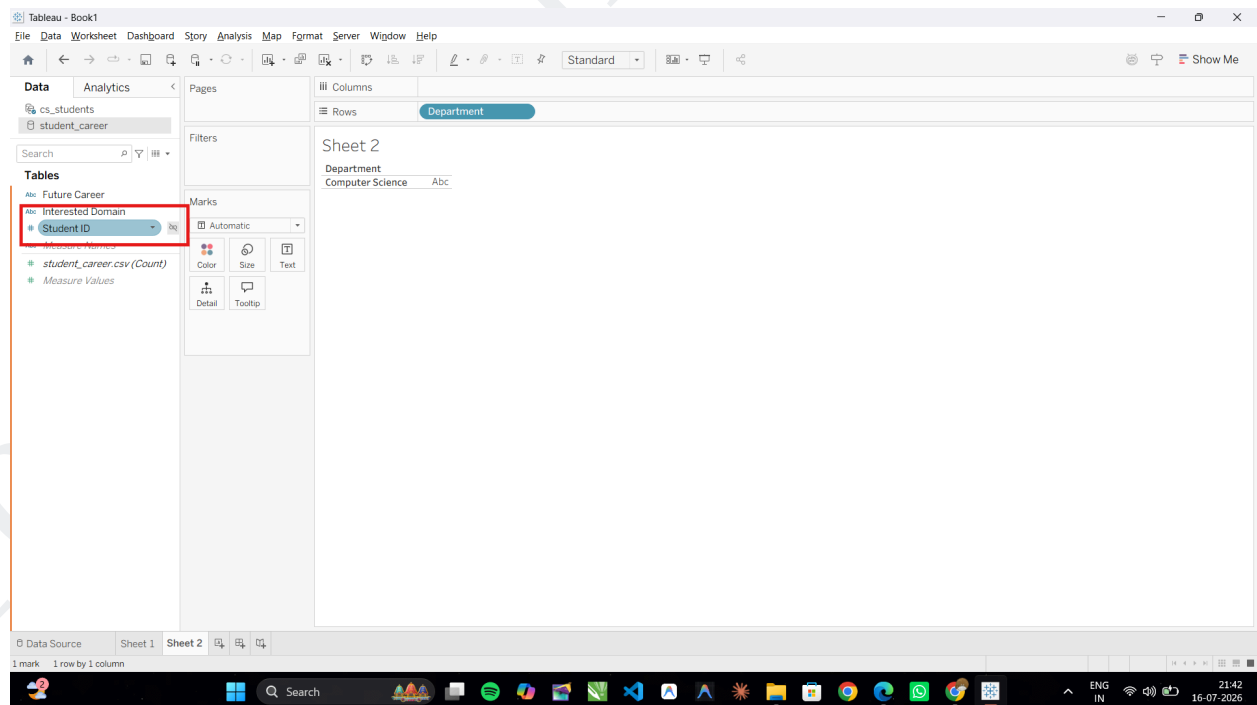
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7. Click **Custom** → **Add** → **OK**

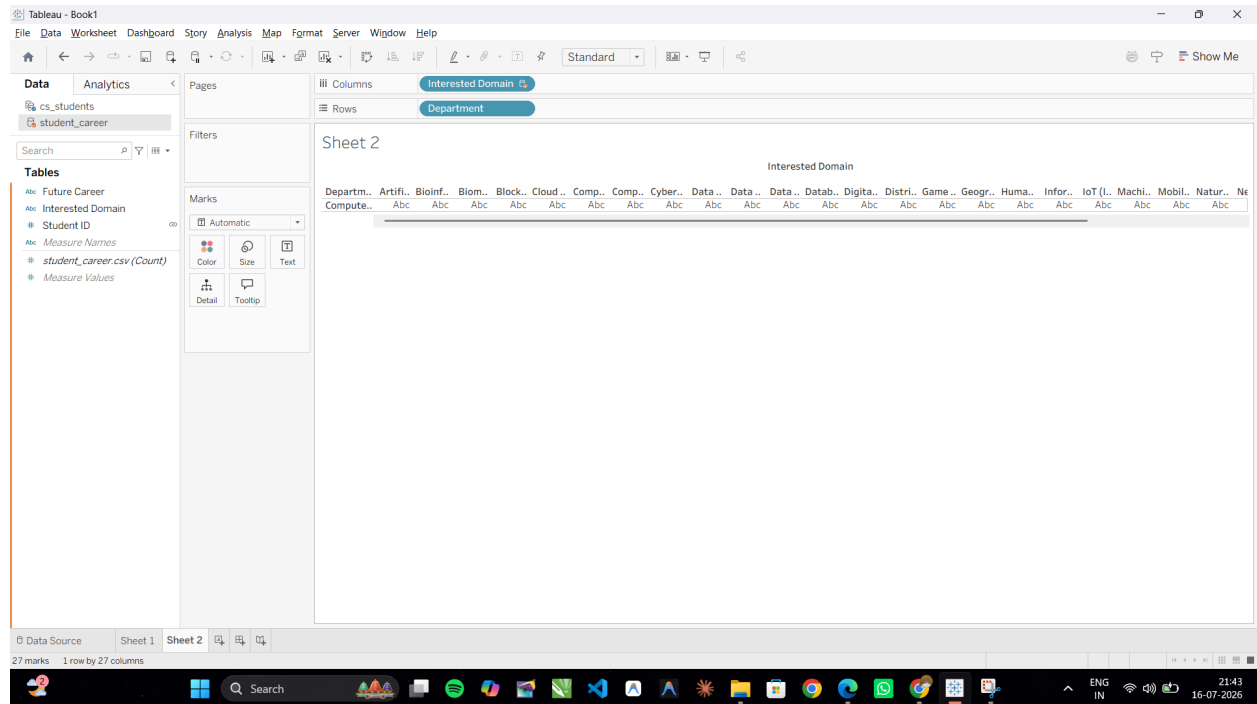


8. Activate the blend relationship by enabling the **Student ID** link field if required.



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9. Drag **Interested Domain from the secondary data source to verify the blend.**

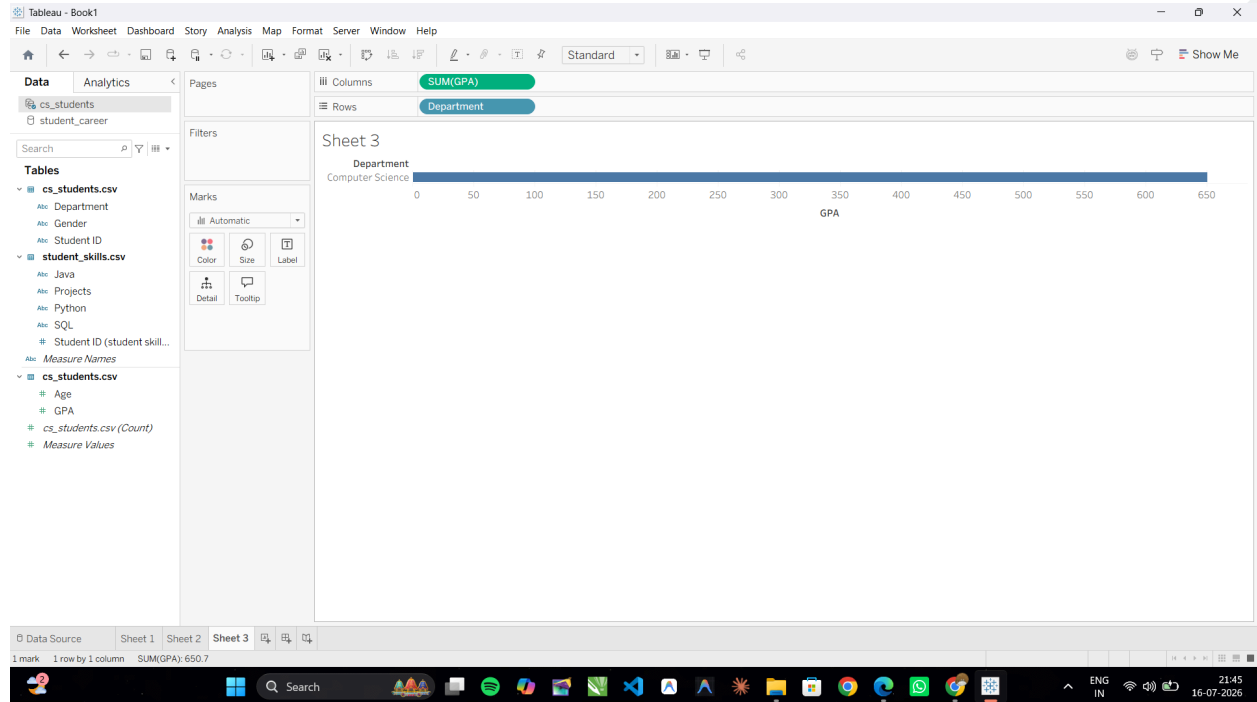


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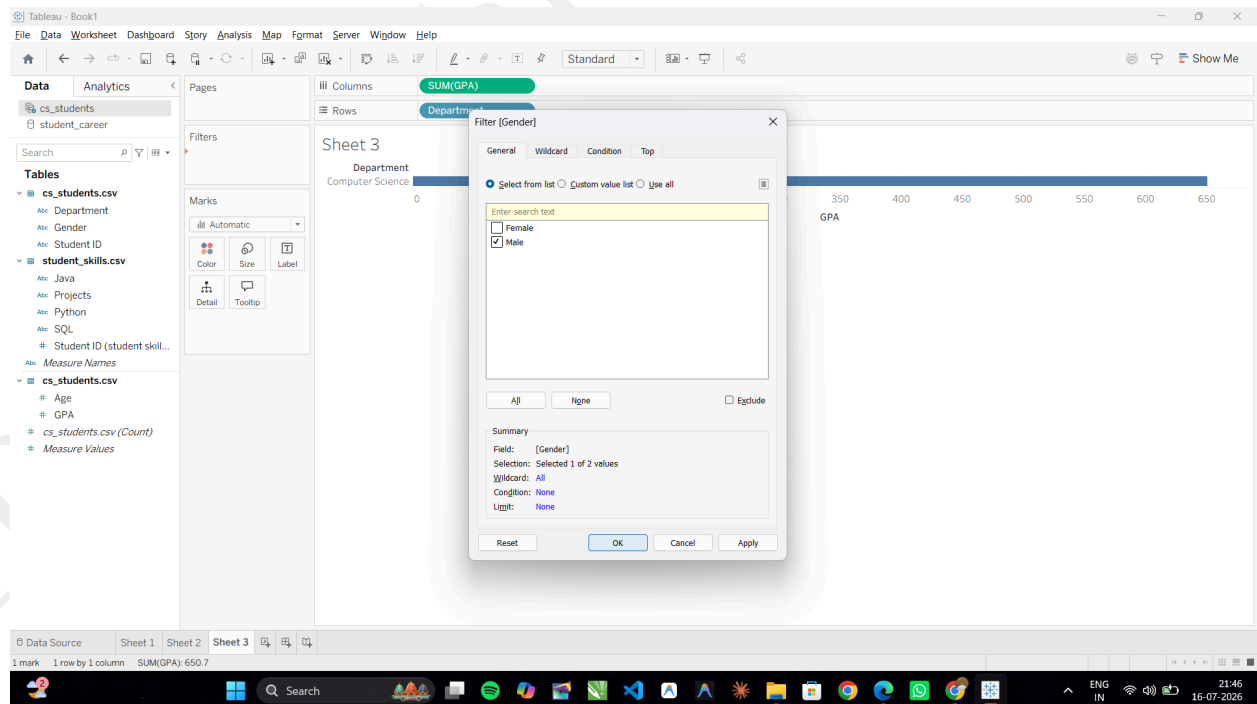
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Step F: Implementing Different Types of Filters

1. Open a **new worksheet**. Drag **Department** to Rows, Drag **GPA** to Columns.



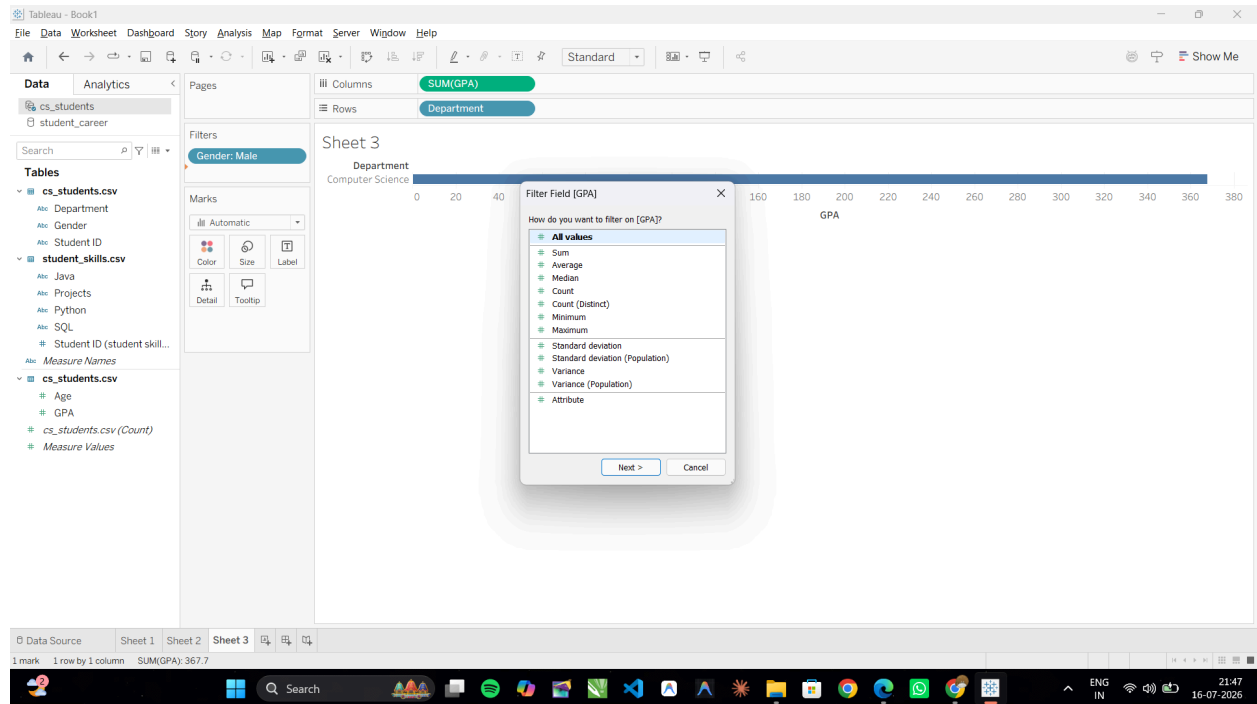
2. Drag **Gender** to the **Filters** shelf Select only **Male** .



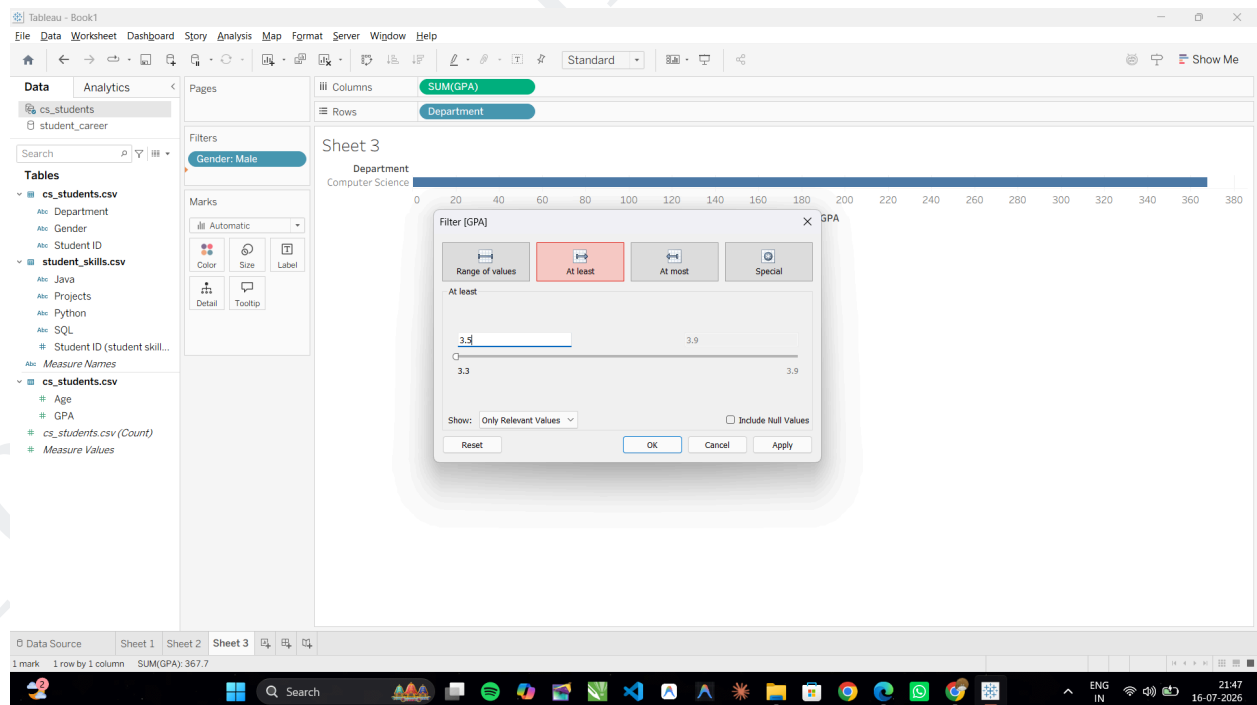
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3. Drag GPA to the **Filters** shelf Select **All Values**, Click **Next**.

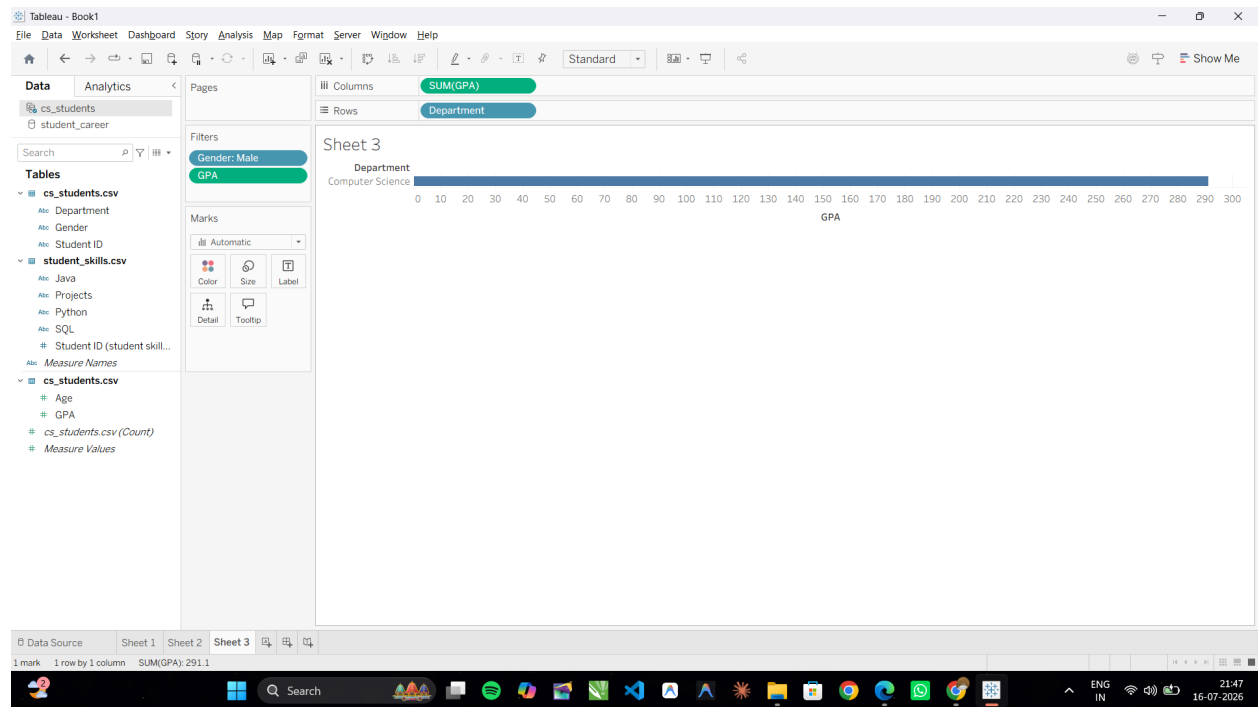


3. Select **At Least**, Enter **3.5**, Click **OK**.



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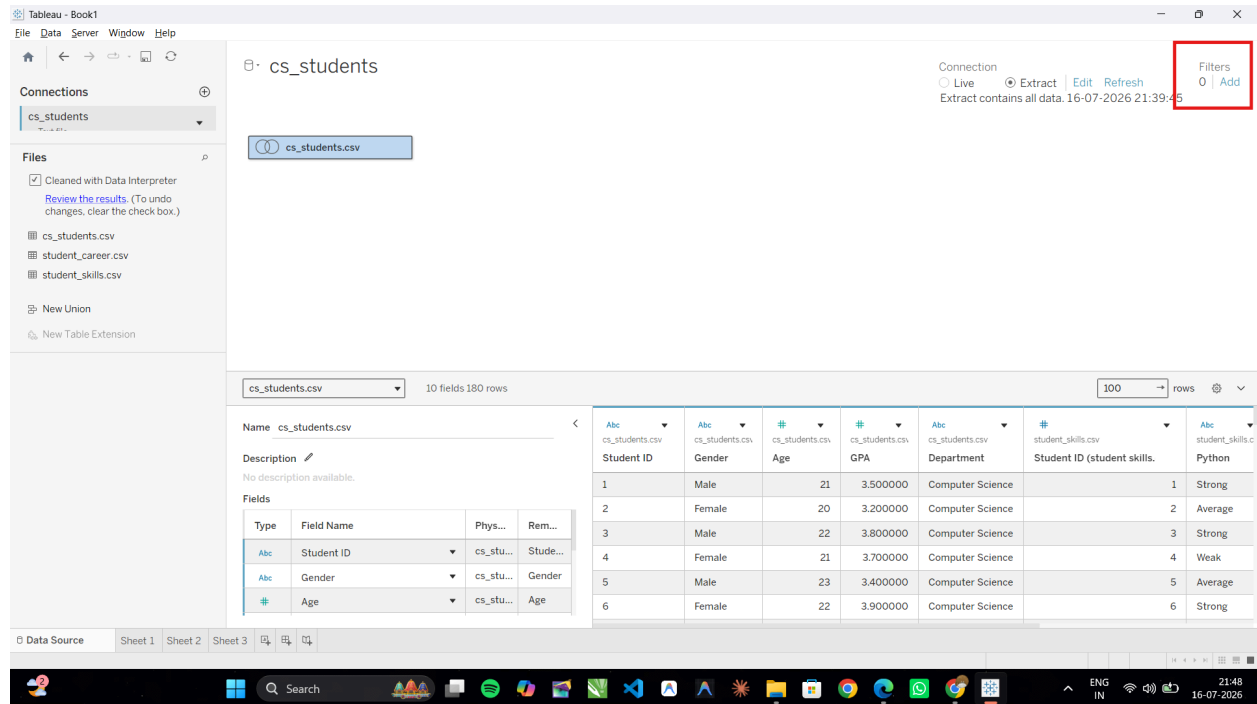
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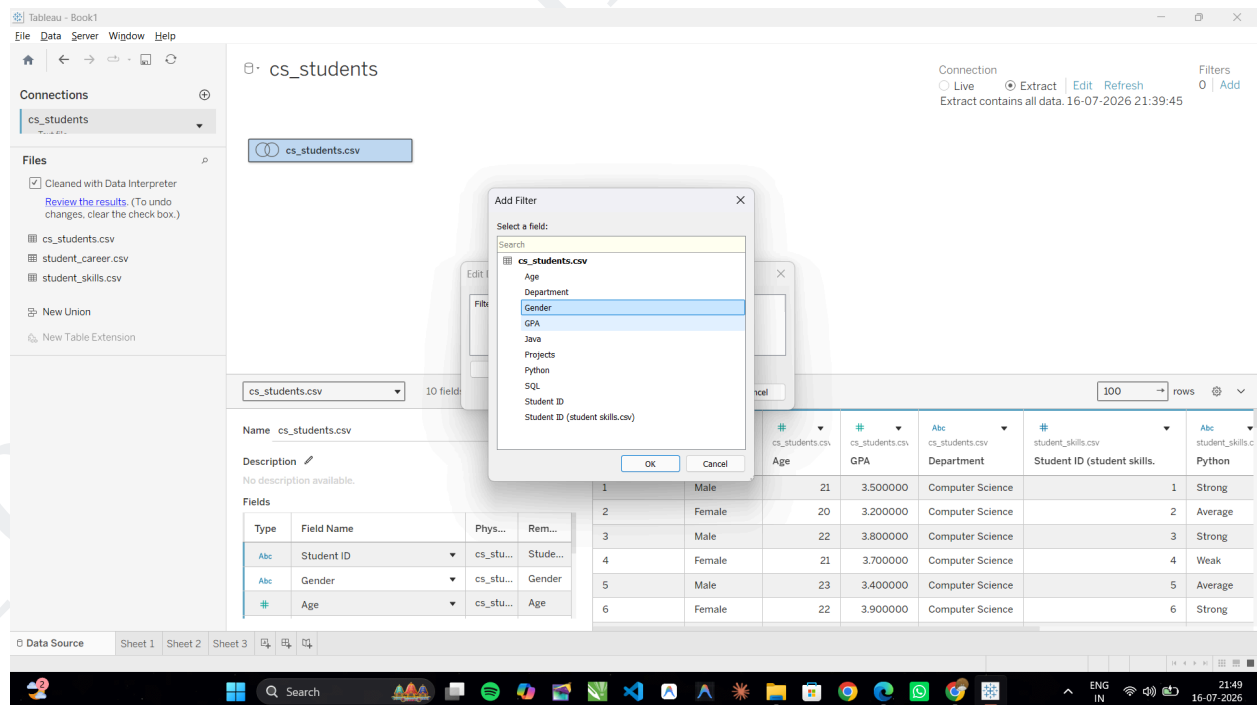
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Step G: Optimizing Dashboard Performance

1. Return to the **Data Source** page.



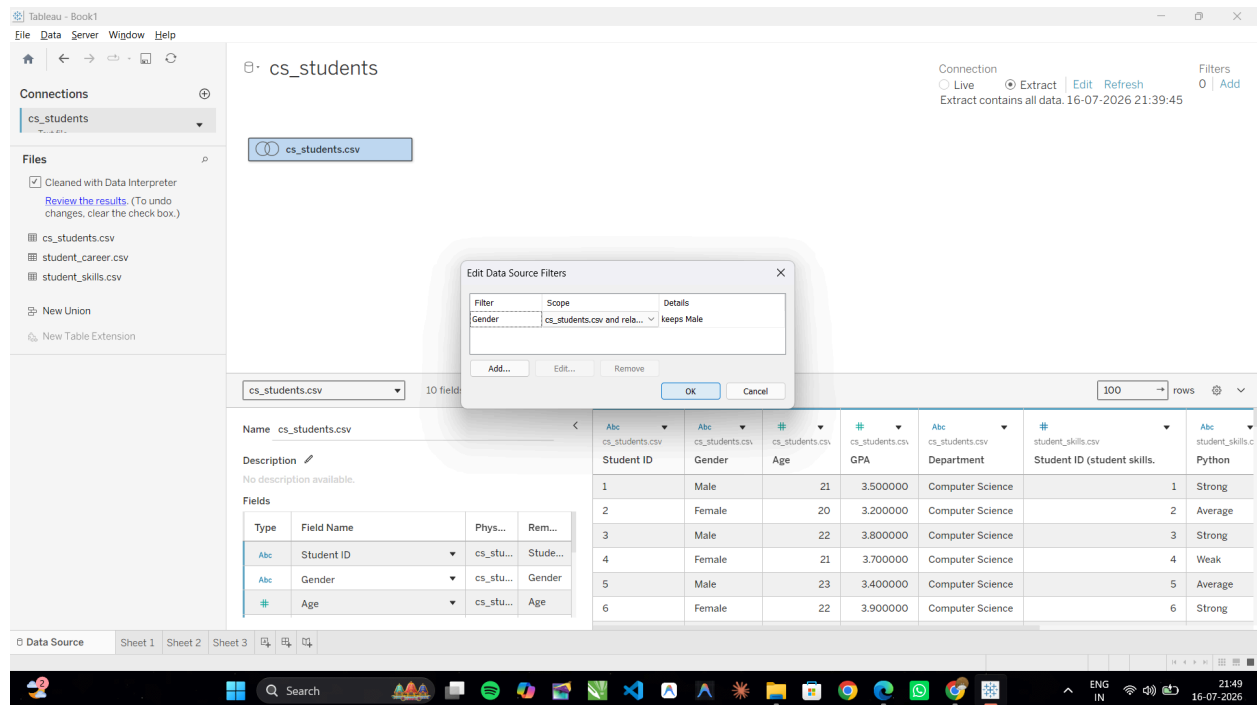
2. Under the **Extract** option, click **Edit**, then click **Add** and Select **Gender** and Choose **Male**.



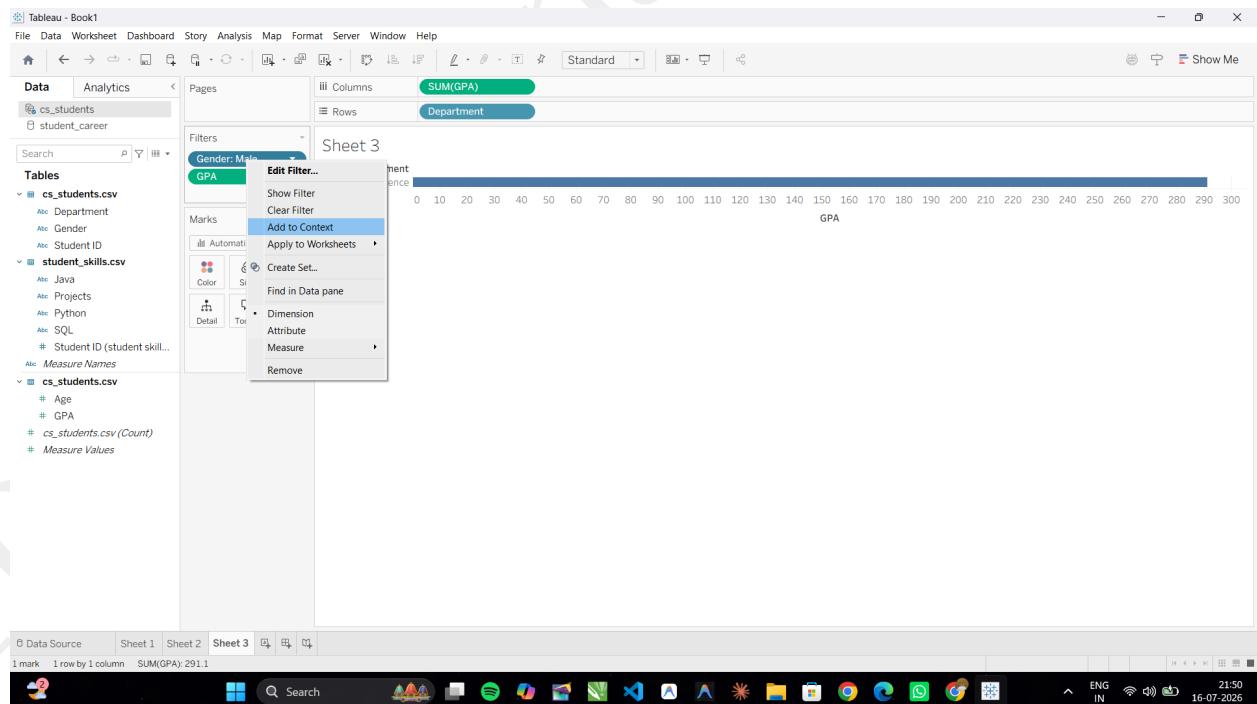
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3. Click OK.



4. Context Filter: Return to the worksheet, Right-click the **Gender** filter on the **Filters** shelf, Select **Add to Context**



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5. The filter pill changes to **gray**, indicating that it has become a **Context Filter**.

